BUSINESS PROCESS EXPLANATION

ROAD_TRAFFIC_ACCIDENT_ANALYSIS | Oracle Database and Power BI

1. Project purpose

The project collects, organizes, manages and analyzes road traffic accident data. Its aim is to identify accident frequency, severity, contributing factors and locations with higher accident levels. The analysis is intended to support road-safety research and evidence-based recommendations.

2. Main actors and inputs

Actor / input	Role in the process
Project researcher / analyst	Defines the research scope, collects data, checks quality, performs analysis and documents findings.
Data providers	Provide accident records and road-safety indicators, such as collision, vehicle, casualty and country-level statistics.
Oracle Database	Stores structured, validated data in related tables and supports SQL queries.
Power BI	Connects to prepared data and presents results in interactive charts and dashboards.

Step 1 – Define the study

Confirm the project objectives, research questions, countries, time period and measures to be analyzed. The planned geographical focus is Rwanda, Kenya, Uganda and other countries where suitable data is available.

Step 2 – Collect data

Obtain relevant datasets from reliable sources. The original project files include collisions.csv, vehicles.csv and casualties.csv. Additional country-level road-safety indicators may be collected from organizations such as WHO and the World Bank.

Step 3 – Check and document sources

For each dataset, record its provider, URL, country coverage, years covered, date obtained, definitions and known limitations. Check whether records describe individual crashes or aggregated national statistics.

Step 4 – Validate and clean data

Inspect column names and data types; identify missing values, duplicates, invalid dates, inconsistent categories and out-of-range values. Keep original files unchanged and save cleaned copies separately. Document any exclusions or corrections.

Step 5 – Prepare data

Standardize formats and category values, define unique identifiers where appropriate, and prepare the fields required for analysis. Do not combine datasets with different levels of detail without a clear method and documented assumptions.

Step 6 – Design the database

Identify entities, attributes, primary keys and foreign keys. Prepare the entity-relationship diagram, relational schema and normalization documentation before creating the Oracle tables.

Step 7 – Implement and load in Oracle

Create tables, constraints and any required views or procedures. Import the validated data and check that row counts, key relationships and data types are correct.

Step 8 – Analyze using SQL

Write SQL queries to summarize accident counts, trends over time, severity, vehicle involvement, casualties and locations. Compare countries only where the source data definitions and coverage are sufficiently comparable.

Step 9 – Present findings

Use Power BI to create visual summaries such as total accidents, accident trends, severity breakdowns, causes or contributing factors, and country comparisons. Prepare a final report with findings, limitations and practical road-safety recommendations.

3. Expected outputs

• Documented and cleaned datasets.

• Database analysis documents, including entities, attributes and relationships.

• An ER diagram and relational schema.

• An implemented Oracle database with tested tables and SQL scripts.

• SQL analysis results and a Power BI dashboard.

• A final report, presentation and evidence of testing.

4. Important data-quality and scope notes

The three CSV filenames indicate the intended data categories, but their exact columns must be checked before the database schema is finalized. For example, the UK STATS19 collision, vehicle and casualty files describe Great Britain; they should not be labelled as Rwanda, Kenya or Uganda data. Country coverage must be stated accurately for every source. Individual accident records and national-level statistics should be analyzed separately unless they can be validly linked and compared.

5. Business process summary

The process starts with defining the research scope and ends with reporting evidence-based findings. Data quality checks take place before loading information into Oracle. Oracle SQL performs the structured analysis, while Power BI communicates selected results visually. The process is iterative: if validation or testing identifies a problem, the relevant data or database step is reviewed and corrected.

Prepared as a project planning document. Update the descriptions after inspecting the actual datasets and confirming the sources used.